HEATMAP NODE · STEM CHECK

HeatMap Node STEM check

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	HMN-STM-001	Version	0.1
Status	Draft	Date	2026-11-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/01b-stem-check.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First STEM check, written at the kit 1.9.1 upgrade of a design that already had a calculation note (HMN-CAL-001); every row recomputed from its governing law in docs/04-calcs/stem_check.py; three detail failures fixed under the standing instruction (HMN-DEC-001 K1, K1b, K2) and one sent to Amish (O4); independent review recorded and applied

HeatMap Node STEM check

Verdict: the working principle of option A (a fan-aspirated radiation shield for air temperature and humidity, a 150 mm black globe for radiant heat, a cup anemometer for wind, and a computed outdoor WBGT) sits within known science, as does the working principle of option B (a directly measured natural wet-bulb temperature on a wetted wick with the same globe). Of 40 rows, **21 are within known science, 15 are plausible, to confirm with measurement, and 4 failed as first written, so a fix is proposed**. Three were fixed under Amish's standing instruction of 2026-10-23: the reservoir of option B splits when frozen if filled to the brim (K1), it runs dry in 7 days in a hot dry spell (K1b), and two nodes cannot resolve a 0.5 C or 1 C urban heat island difference from single readings (K2, a claim restated, not a design change). The fourth, S25 (the FieldNode cell charging in a 57.2 C interior), belongs to the FieldNode repo and goes to Amish with the fix proposed (HMN-DEC-001 O4). The independent review (below) also downgraded the shield claim S6 to plausible, which puts PR-001 at risk, and corrected the arm check (S9; a minimum yield of 110 MPa and the weight of the head added: factor 7.5 in place of 13.0). None of the fixes changes the working principle, the pitch or the scope. `stem_check` stays **open** because 15 rows can be closed only by measurement. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which imports `docs/04-calcs/sizing.py` (HMN-CAL-001) and computes each row from its own law rather than copying the note's result.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`). Rows S31 to S37 concern option B only (marked [B]); the rows with the suffix b are the claims after a fix.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Worked example: a 50 C globe in 35 C air at 1 m/s sits in surroundings of MRT 74.8 C	Physics	Heat transfer	Steady globe balance eps sigma (Tmrt^4 - Tg^4) = h (Tg - Ta), forced h = 6.3 v^0.6 / D^0.4 (ISO 7726 form)	in 202.2 W/m2, out 201.8 W/m2	Within known science	HMN-CAL-001 A1
S2	On a hot paved street in full sun the mean radiant temperature stays at or below 80 C	Physics	Optics and radiation	Stefan-Boltzmann with sphere view factors: ground 65 C, sky 25 C effective, 900 W/m2 direct and 120 W/m2 diffuse sun, absorptivity equal to emissivity	79.8 C <= 80 C	Plausible, to confirm at TRL 3	HMN-CAL-001 I1 (sizing case)
S3	The 150 mm black globe reaches about 67 C at 50 C air, MRT 80 C and 0.3 m/s	Physics	Heat transfer	Globe balance solved for Tg with the larger of forced and natural convection coefficients	claimed 67 C, estimate 67.4 C (ratio 0.99)	Within known science	HMN-CAL-001 I1
S4	Globe 90 % response is within 30 min (about 12 min with the probe bead)	Physics	Heat transfer	Lumped capacitance tau = m c / (h A) of the copper shell; 90 % response 2.3 tau; the 10 mm bead adds a lag of about 4 min (CAL D5); conduction along the probe stem not included	11.5 min <= 30 min	Within known science	HMN-CAL-001 D2, D5
S5	Psychrometric wet bulb of 35 C, 40 %RH air is 24.2 C	Physics	Thermodynamics	Stull (2011) empirical fit of psychrometric wet bulb, Pa 101.3 kPa	claimed 24.2 C, estimate 24.5 C (ratio 0.99)	Within known science	HMN-CAL-001 B1
S6	With the fan the shield reads 0.43 C high in full sun at wind speeds of 1 m/s and more	Physics	Heat transfer	Energy balance: every watt the plates absorb (3.74 W) could at most warm the 4.0 L/s fan flow by absorbed / (rho cp Q) = 0.79 C, above the 0.5 C target; the model value 0.43 C	Plausible only: bound 0.79 C against the 0.5 C target; fan flow, sensor lag and plate-to-air coupling to measure (PR-001 now at risk)	Plausible, to confirm at TRL 3	HMN-CAL-001 C4

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				(HMN-CAL-001 C4) rests on assumed 50 % heat transfer to the air stream, a 2 s sensor lag and a 6 s fan run, and it gets worse below 0.5 m/s (independent review: SHT45 lag on a board under a cap may be 5 to 10 s, giving 0.6 to 0.75 C)			
S7	The 60 mm fan draws 4.0 L/s through the plate stack (60 % of its 6.6 L/s free-air rating)	Physics	Fluid mechanics	Continuity: 1.6 m/s up the 56 mm column is 3.9 L/s; 14 CFM is 6.6 L/s; the stack's inlet area (110 mm x 12 mm gaps x 7) keeps the dynamic head near 1.5 Pa, far below a 60 mm fan's static pressure (class values, not a datasheet)	Fan curve of the chosen 60 mm fan against the stack not yet known	Plausible, to confirm at TRL 3	Open
S8	A 35 m/s gust gives 750 Pa and 6.6 N on the globe	Physics	Fluid mechanics	Dynamic pressure 0.5 rho v^2 = 750 Pa; drag on a sphere Cd 0.47 (subcritical)	claimed 6.6 N, estimate 6.23 N (ratio 1.06)	Within known science	HMN-CAL-001 H1
S9	The arm stays below yield / 5 at a 35 m/s gust (wind 11 MPa, with the weight of the head 14.7 MPa, against the 110 MPa minimum yield)	Physics	Strength of materials	Cantilever bending sigma = M / Z, 25 x 25 x 2 mm tube; wind and gravity bending added at the section corner (conservative); 6063-T5 minimum yield about 110 MPa (typical 145 MPa, stockist certificate to confirm); required factor 5; bolt holes	14.7 MPa <= 22 MPa	Within known science	HMN-CAL-001 H2, H2b

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				at the root not checked			
S10	The arm tube bends less than 10 mm at its tip at 20 m/s (tube bending only; rotation in the clamp and gravity sag not included)	Physics	Strength of materials	Cantilever deflection F L^3 / (3 E I), E = 69 GPa, rigid root assumed	0.29 mm <= 10 mm	Within known science	HMN-CAL-001 H3 (0.31 mm)
S11	Two strap bands at 1,000 N preload hold the arm against a twist of 18 N m (factor 2.5)	Physics	Mechanics and statics	Band friction on the pole with the lined V: torque = n mu T r = 46 N m for mu 0.4 (FND-CAL-001); the preload and the rubber-on-pole friction are assumed	Preload and friction not measured	Plausible, to confirm at TRL 3	Open
S12	The fan, run 6 s every 3 min, adds 35.3 mW of average load	Physics	Electromagnetism and circuits	Average power = P x duty / efficiency	claimed 35.3 mW, estimate 35.3 mW (ratio 1.00)	Within known science	HMN-CAL-001 C5, F2
S13	Sensors and fan (35.4 mW) stay inside FieldNode's 100 mW sensor allowance, so the 5.75 days without sun of FND-CAL-001 hold	Physics	Electromagnetism and circuits	Energy balance of the 3.2 V 6 Ah LiFePO4 cell (19.2 Wh) at the allowance, FND-CAL-001	35.4 mW <= 100 mW	Within known science	HMN-CAL-001 F2; FND-CAL-001
S14	A 10 kOhm NTC (B 3950) reads 3.59 kOhm at 50 C	Engineering and technology	Sensing resolution	Beta equation R = R25 exp(B (1/T - 1/298.15))	claimed 3.59 kOhm, estimate 3.59 kOhm (ratio 1.00)	Within known science	HMN-CAL-001 D3
S15	A 20 byte payload every 15 min costs 23.7 s of airtime a day at SF9	Physics	Electromagnetism and circuits	LoRa time on air (Semtech AN1200.13) with 13 bytes of LoRaWAN overhead, 125 kHz, CR 4/5	claimed 23.7 s/day, estimate 23.7 s/day (ratio 1.00)	Within known science	HMN-CAL-001 G2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S16	One uplink (247 ms) is shorter than the 400 ms dwell limit on a US915 channel	Engineering and technology	Standards practice	FCC 47 CFR 15.247 and LoRa Alliance RP002 regional parameters (US902-928: 400 ms dwell, 125 kHz uplink channels)	247 ms <= 400 ms	Within known science	HMN-REG-001 G3
S17	Reed-contact pull-up worst case is 0.109 mW	Physics	Electromagnetism and circuits	P = V^2 / R, 3.3 V across 100 kOhm	0.109 mW <= 0.11 mW	Within known science	HMN-CAL-001 F1
S18	The forced-convection globe coefficient 6.3 / sigma equals the ISO 7726 constant 1.1e8	Mathematics	Units and dimensional analysis	Unit conversion between h (W/m2K) and the MRT form (K^4 per ...)	claimed 1.1e+08 , estimate 1.11e+08 (ratio 0.99)	Within known science	HMN-CAL-001 A1
S19	Input errors add up to 0.30 C of WBGT as a root sum of squares of random terms (a warm shield bias of up to 0.43 C and a height bias of up to 0.4 C are one-sided and come on top)	Mathematics	Statistics and uncertainty	Quadrature sum of the four contributions 0.23, 0.15, 0.09, 0.09 C; independence assumed, SHT45 typical figures, Liljegren model error not included	0.3 C <= 0.31 C	Within known science	HMN-CAL-001 B4
S20	Daytime urban heat island differences of 0.5 to 4 C (EPA: 1 to 7 F) can be seen with a node error of 0.43 C	Mathematics	Statistics and uncertainty	Unit conversion 1 F = 0.556 C; two nodes with independent random errors of 0.43 C differ by 0.61 C in quadrature, so a single reading resolves a difference only at about 1.2 C (95 % %); a sunny site also reads up to 0.43 C warm	Fails for 0.5 C and 1 C; fix K2: restate the claim to differences of 2 C or more between single readings of sunny and shaded sites (the upper part of the EPA range; means over a season resolve smaller random	Fails, fix proposed	HMN-DEC-001 K2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				against a shaded one	differences but not the sun bias)		
S20b	After fix K2, differences of 2 C or more between single readings are resolved: 1.96 sigma of 0.61 C plus the 0.43 C sun bias	Mathematics	Statistics and uncertainty	95 % half-width of the difference of two independent 0.43 C random errors (1.96 x 0.61 C) plus the one-sided 0.43 C shield bias	1.62 C <= 2 C	Within known science	HMN-DEC-001 K2
S21	WBGT of 31.7 C follows from the formula 0.7 Tnwb + 0.2 Tg + 0.1 Ta	Mathematics	Units and dimensional analysis	Arithmetic of ISO 7243:2017 (outdoors with solar load): 0.7 x 26.0 + 0.2 x 50 + 0.1 x 35; the 26.0 C wet bulb of the note applies emissivity 0.95 to the whole radiant load of a white wick and so reads high (conservative)	claimed 31.7 C, estimate 31.7 C (ratio 1.00)	Within known science	HMN-CAL-001 B2
S22	316 stainless fixings and bands in contact with 6063 and 5052 aluminium survive 10 years of weather	Chemistry	Electrochemistry and batteries	Galvanic couple of stainless steel on aluminium in a wet atmosphere: the aluminium is the anode and corrodes at the crevice; the cheeks, sleeves and saddle are separated from the stainless bolts by 8 mm aluminium crush sleeves and EPDM trim only in part	Corrosion at the bolt holes in salt air is not ruled out; sealant or plastic washers would reduce it	Plausible, to confirm at TRL 3	Open
S23	White ASA plates, cowl and adapter blocks keep their shape and colour at the	Chemistry	Materials and polymer chemistry	ASA is a UV-stable ABS-family polymer with a glass transition near 100 C (supplier datasheets; the filament is not	Heat deflection temperature of the chosen filament to confirm	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	hottest shield temperature			chosen); shield plates run about 15 to 25 K above air in the sun, so 65 to 75 C at 50 C air			
S24	The matte black globe paint holds its emissivity of about 0.95 as it ages	Chemistry	Materials and polymer chemistry	Matte black paints have emissivity of about 0.95 at the start; fading and dust change it over years	Ageing unknown; recalibrate in CalRig	Plausible, to confirm at TRL 3	Open
S25	The FieldNode LiFePO4 cell keeps charging in the shaded enclosure at 50 C air (57.2 C interior, FND-CAL-001)	Chemistry	Electrochemistry and batteries	LiFePO4 cells are usually specified to charge between 0 and 45 C and discharge to 60 C (cell datasheets vary); the interior of 57.2 C at 50 C air is above the usual charge limit	Fails in a heat wave: no charging above the limit and faster ageing; fix proposed to Amish and FieldNode (charge cut-off by cell temperature, HMN-DEC-001 O4, PR-016); not made here because the core belongs to the FieldNode repo	Fails, fix proposed	HMN-DEC-001 O4
S26	WBGT, derived from the node's readings, is a screening index for heat strain, not a judgement on any one person	Biology	Human physiology and biomechanics	ISO 7243:2017 defines WBGT as a screening index; NIOSH (2016) criteria use it with work rate and acclimatization; the node reports environmental values only (planning and research data, not a heat warning)	Within the standard's stated use	Within known science	HMN-PRC-001 Safety
S27	Wasps, spiders and dust do not block the shield gaps or the fan	Biology	Ecology and agronomy	Insects nest in sheltered gaps of radiation shields (a known field problem for	Field observation needed; inspection at each visit	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				naturally ventilated shields); 12 mm gaps, a fan every 3 min and a cowl lid reduce the risk			
S28	The Sensirion SHT45 holds +/-0.1 C and +/-1.0 %RH (typical) and survives 67 C in the shield	Engineering and technology	Sensing resolution	Sensirion SHT45 product page: typical +/-0.1 C and +/-1.0 %RH; operating range -40 to 125 C; typical values apply at 25 C, not over the whole range	Within the datasheet's stated typical figures; maximum values are wider	Within known science	HMN-CAL-001 B3
S29	The cup anemometer starts at about 0.8 m/s	Engineering and technology	Sensing resolution	Typical low-cost three-cup pulse anemometers start between 0.5 and 1 m/s (maker data, unit not chosen)	Start-up not measured; the calm share is reported and flagged	Plausible, to confirm at TRL 3	Open
S30	The sensor head (without the FieldNode core) weighs under 4 kg: 2.11 kg in HMN-CAL-001 K1 (the D3 model of made and drawn parts sums to 1.60 kg without catalogue masses of bought parts)	Engineering and technology	Structural and mechanical design	Sum of component masses in sizing.py (volumes of the D3 model at each part's material density, bought parts at catalogue masses)	2.11 kg <= 4 kg	Within known science	HMN-CAL-001 K1, K2
S31	[B] The 56 mm reservoir keeps the wick wet for 7 days at 45 C, 15 %RH, 1 m/s and MRT 80 C, with a 15 % headspace	Physics	Heat transfer	Wick energy balance (sizing.tnwb) gives wet bulb 25.5 C and an evaporation flux of 0.48 g/m2 s over 7.5 cm2 of sleeve, 31.5 g a day if the worst case lasted 24 h; 7 days need 221 mL against 186 mL	Fails by 19 % in the extreme case; fix K1b: refill interval of 5 days in hot dry spells (158 mL) and 7 days otherwise	Fails, fix proposed	HMN-DEC-001 K1b

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				usable at an 85 % fill			
S31b	[B] After fix K1b (the wick cord enclosed in a sleeve, so only the 7.5 cm2 sensor sleeve evaporates) the reservoir keeps the wick wet for 5 days at the same extreme	Physics	Heat transfer	Same wick balance; 5 days at 31.5 g a day against 186 mL usable (margin 15 %, sensitive to wind: flux grows as v^0.47); an open 3 mm cord over the 0.12 m lift would add about 11 cm2 and more than double the use	Plausible: the sleeve must be sealed from the cord's neck to the sensor; delivery rate (S33) to measure	Plausible, to confirm at TRL 3	HMN-DEC-001 K1b
S32	[B] A cotton wick lifts water the 0.12 m from the reservoir surface to the sensor	Physics	Fluid mechanics	Jurin's law h = 2 gamma cos(theta) / (rho g r) for 20 micrometre pores, gamma 0.072 N/m, complete wetting: 0.73 m static lift	0.12 m <= 0.73 m	Within known science	HMN-OPT-001
S33	[B] The wick delivers water as fast as the sensor sleeve evaporates it (0.37 mg/s) at 1 m/s	Physics	Fluid mechanics	Darcy flow through the 3 mm cotton cord needs the cord's permeability, which is not known here	Delivery rate to measure with the first wick	Plausible, to confirm at TRL 3	Open
S34	[B] The reservoir bottle, filled to the brim, survives the water freezing at -20 C (R8)	Chemistry	Physical chemistry (kinetics, equilibria, thermochemistry)	Water expands 9.07 % on freezing; a brim-full bottle (3 % headspace) is split or its neck pushed off	Fails with 3 % headspace against 9.07 % expansion; fix K1: 85 % fill line and an invalid-below-5 C flag on the wet-bulb channel	Fails, fix proposed	HMN-DEC-001 K1
S34b	[B] After fix K1 the bottle survives the water freezing at -20 C: 85 % fill (15 % headspace) against 9.07 % expansion, a	Chemistry	Physical chemistry (kinetics, equilibria, thermochemistry)	Water expands 9.07 % on freezing; headspace is necessary but not sufficient (ice forms first at the wall and neck and can trap liquid); the loose gland gives the ice	Plausible: freeze behaviour of the bottle to test at TRL 4	Plausible, to confirm at TRL 3	HMN-DEC-001 K1

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	thin HDPE wall and a gland that is a loose push fit, so ice pushes it out			a path; the invalid-below-5 C flag only protects the data			
S35	[B] Distilled water and a clean wick keep the wet-bulb reading true for the visit interval	Chemistry	Analytical chemistry	Evaporation concentrates dissolved salts on the wick and slows evaporation (the reason natural wet-bulb practice calls for distilled water and clean wicks); the interval is unknown	Wick life in street dust to measure; monthly visit assumed	Plausible, to confirm at TRL 3	Open
S36	[B] A closed reservoir with a 6 mm gland breeds no mosquitoes or algae	Biology	Microbiology	Container mosquitoes lay in open water of a few centimetres; the cap has no open surface, only the sensor gland and a wick; light-tight dark HDPE limits algae; stagnant water in a closed bottle can still grow biofilm	Field observation needed	Plausible, to confirm at TRL 3	Open
S37	[B] The directly measured WBGT is within +/-1.0 C (R5): 0.7 x wick 0.5 C, 0.2 x globe 0.3 C, 0.1 x passive shield 1.04 C	Mathematics	Statistics and uncertainty	Quadrature sum 0.37 C, linear sum 0.51 C; the 0.5 C wick error is assumed and is the largest term; wick fouling (S35) and non-standard wick geometry not included	Plausible: wick error to measure against a reference WBGT meter	Plausible, to confirm at TRL 3	HMN-OPT-001 Table 1

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**.

Requirement links: S25 is product requirement PR-016 (cell charge limit); S27, S31 and S31b bear on PR-018 (upkeep); S2, S5, S19 and S21 on PR-005 (WBGT); S6 on PR-001 (air temperature); S9 on PR-011 (wind).

Fixes made under the standing instruction

Item	What failed	Fix (HMN-DEC-001)	Result
Reservoir freezing (option B, S34)	A brim-full bottle has 3 % headspace against the 9.07 % expansion of freezing water, so at -20 C (PR-008) the bottle splits or its neck is pushed off	K1: fill line at 85 % of the bottle (15 % headspace) marked in the bottle and in the maintenance steps, a thin HDPE wall, a gland that is a loose push fit so ice pushes it out rather than splitting the bottle, and a firmware flag that marks the wet-bulb channel invalid below 5 C	9.07 % expansion against 15 % headspace; plausible (S34b), freeze behaviour to test at TRL 4
Reservoir capacity (option B, S31)	At 45 C, 15 %RH, 1 m/s and MRT 80 C the wick sleeve evaporates 31.5 g a day, so the 186 mL usable at an 85 % fill lasts 5.9 days, short of the 7 days claimed	K1b: refill interval of 5 days in hot dry spells and 7 days otherwise (PR-018), and the wick cord enclosed in a sealed sleeve from the neck to the sensor so that only the 7.5 cm2 sensor sleeve evaporates (an open cord would add about 11 cm2 and shorten the interval to about 2 days, independent review)	158 mL in 5 days against 186 mL usable; plausible (S31b), delivery rate to measure (S33)
Heat island resolution (S20)	The claim that a 0.5 C urban difference can be seen overstated the node: two independent 0.43 C random errors differ by 0.61 C in quadrature, a single reading resolves only about 1.2 C at 95 %, and a sunny site reads up to 0.43 C warm against a shaded one (independent review)	K2: the claim is restated to differences of 2 C or more between single readings (the upper part of the EPA range of 0.5 to 4 C); means over a season resolve smaller random differences but not the sun bias	1.62 C needed against 2 C (S20b)
Cell charge limit (S25)	The FieldNode interior of 57.2 C at 50 C air is above the usual 45 C charge limit of LiFePO4, so the cell would not charge in a heat wave	Proposed, awaiting Amish (O4, PR-016): a charge cut-off by cell temperature in FieldNode's firmware; the core belongs to the FieldNode repo, so nothing is changed here	Open

K1, K1b and K2 do not change the working principle, the pitch or the scope. They are recorded in HMN-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse them. K1 and K1b apply to option B only; Amish has not chosen between A and B, and the choice stays his (HMN-OPT-001, HMN-DDR-004).

Fields not relevant

None. Every field is touched: physics (globe and wick energy balance, radiation, strength of materials, wind load, circuit loads), chemistry (corrosion, polymers, cell temperature limits, freezing), biology (heat stress screening index, insects and algae), mathematics (uncertainty and units) and engineering (sensing, radio and manufacturing).

Concept options

- Option A, fan-aspirated shield with a globe and anemometer: the rows S1 to S30. Within known science in principle; the two unmeasured assumptions that matter most are the fan flow through the plate stack (S7) and the preload of the strap bands (S11).

- Option B, direct natural wet-bulb head: rows S31 to S37. Within known science in principle (a wetted wick in natural wind is the ISO 7243 reference measurement of the natural wet-bulb temperature); the details that failed were the freezing and capacity of the reservoir (fixed by K1 and K1b, both still plausible after the fix). The wick's delivery rate (S33), distilled-water upkeep (S35), mosquito and biofilm control (S36) and the 1.0 C accuracy of the measured WBGT (S37) are plausible, to confirm in the field.

Independent review

A second reviewer, a headless Claude session started with `claude -p --model claude-opus-5-5` (the current model, as Amish set on 2026-10-28) in an empty folder holding only this document, the calculation note HMN-CAL-001 (`docs/04-calcs/01-sizing.md`), the product requirements HMN-PRS-001 and the script `docs/04-calcs/sizing.py` , and none of the author's reasoning, was asked to refute the headline results and every "Within known science" verdict. It recomputed the rows with its own code on 2026-11-01. Its findings and what changed:

Row	Finding	Change
Counts, S1, S3, S8, S12, S14 to S18, S26, S28, S32	Upheld (S1's balance check is circular, because the MRT came from the same formula; S8's Reynolds number is at the drag crisis, so a drag coefficient of 0.47 is a reasonable midpoint, not clearly conservative)	None
S4	Upheld; wording wrong: the bead lag is 4.2 min, not 4 tau; conduction along the stem not included	Wording corrected
S5	Upheld; an exact psychrometric solution gives 24.0 C and Stull's fit 24.5 C, against 24.2 C	None
S6	Refuted as worded: the energy-balance bound is 0.79 C, above the 0.5 C target, so 0.43 C rests on assumed plate-to-air coupling (50 %), a 2 s sensor lag and a 6 s run, and "at any wind speed" is false below about 0.5 m/s; ground long-wave and diffuse light are left out	Downgraded to plausible; claim limited to 1 m/s and more; PR-001 now at risk
S9	Weakened: gravity bending was left out and 145 MPa is a typical, not a minimum, yield (6063-T5 about 110 MPa); bolt holes at the root not checked	sizing.py H2 uses 110 MPa (factor 9.9) and H2b adds gravity (14.7 MPa, factor 7.5); the verdict stands on the new figures; bolt holes carried as an open item
S10	Weakened: assumes a rigid root; rotation in the clamp and gravity sag not included	Claim restated to tube bending only
S11	The law cell used mu 0.5 (57 N m) while the claim's factor of 2.5 uses mu 0.4 (45.7 N m); creep of the trim and band expansion missing	Script uses mu 0.4; limits stated; plausible
S13	Upheld but inherited from FND-CAL-001; charging in heat (S25) is not in the row	Cross-reference to S25
S19	Arithmetic upheld, claim weakened: the shield and height biases are one-sided and not random; SHT45 typical values; model error excluded; the wick model gives the white wick the long-wave emissivity of 0.95 for the whole radiant load (Liljegren uses a solar absorptivity near 0.6), so the wet bulb and S21's 26.0 C read high	Claim worded as random terms with the biases on top; the wick point recorded as conservative and carried to REVIEW.md, no figure changed
S20b	Refuted: the "0.7 of the difference" test was invented; 1.6 sigma at 1 C; errors not independent because the sun bias is one-sided	K2 restated to 2 C or more with the 95 % half-width plus bias (1.62 C)
S21	Upheld as arithmetic only	Note on the wick bias added
S25	Should be a failure: no charging in the heat waves that matter	Verdict changed to "Fails, fix proposed" and sent to Amish (O4)
S30	Weakened: the row quoted the model's 1.6 kg, not the note's 2.11 kg, and had the wrong field label	Row now quotes sizing.py (2.11 kg)
S31b	Weakened: an open 3 mm cord adds about 11 cm2 of evaporating area and more than doubles the use; margin only 15 %; wind raises the flux	Enclosed cord added to K1b; downgraded to plausible
S34b		

Row	Finding	Change
	Weakened: expansion is 9.07 %, not 9.17 %; headspace is necessary, not sufficient (ice seals the neck and traps liquid); the invalid flag protects data only	9.07 % used; thin wall and loose gland added to K1; downgraded to plausible
S37	Verdict should be plausible: the 0.5 C wick error is a guess; biases added in quadrature (linear sum 0.51 C)	Downgraded to plausible; linear sum stated
Missing	Fan failure or dust (a dead fan silently gives 1 to 2 C errors); rain or dew on the globe; vibration and fatigue of the arm and clamp; wind correction for turbulence on the globe; light reflected from facades beyond the 80 C MRT case; lightning and earthing on a metal pole	Fan fault, rain on the globe and fatigue added to the design FMEA (HMN-RSK-001 F13 to F15); the others recorded in REVIEW.md as open items

The reviewer would also have PR-001 at risk, which is now its status. Nothing was upheld on its own authority: every figure above was recomputed by the reviewer and then by the author.

Sources

- ISO 7243:2017, *Ergonomics of the thermal environment: assessment of heat stress using the WBGT (wet bulb globe temperature) index*, https://www.iso.org/standard/67188.html, accessed 2026-11-01 (S21, S26).
- Liljegren, J. C., R. A. Carhart, P. Lawday, S. Tschopp and R. Sharp, "Modeling the wet bulb globe temperature using standard meteorological measurements," *Journal of Occupational and Environmental Hygiene* 5 (2008): 645 to 652, https://doi.org/10.1080/15459620802310770 (S3, S5, S31).
- NIOSH, *Criteria for a Recommended Standard: Occupational Exposure to Heat and Hot Environments*, Publication 2016-106, https://www.cdc.gov/niosh/docs/2016-106/default.html, accessed 2026-11-01 (S26).
- Stull, R., "Wet-bulb temperature from relative humidity and air temperature," *Journal of Applied Meteorology and Climatology* 50 (2011): 2267 to 2269 (S5).
- Semtech, *AN1200.13 LoRa Modem Designer's Guide* (time on air formula) and LoRa Alliance, *RP002-1.0.4 Regional Parameters*, https://resources.lora-alliance.org/technical-specifications/rp002-1-0-4-regional-parameters, accessed 2026-11-01 (S15, S16).
- FCC, 47 CFR 15.247, https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-C/subject-group-ECFR2a5a6f3d3f3e4a6/section-15.247, accessed 2026-11-01 (S16).
- Sensirion, SHT45 product page, https://sensirion.com/products/catalog/SHT45, accessed 2026-11-01 (S28).
- US EPA, "Learn about heat islands," https://www.epa.gov/heatislands/learn-about-heat-islands, accessed 2026-11-01 (S20).
- WMO, *Guide to Instruments and Methods of Observation* (WMO-No. 8), https://www.wmo.int/pages/prog/www/IMOP/CIMO-Guide.html, accessed 2026-11-01 (radiation shields, S6).
- Young, W. C., and R. G. Budynas, *Roark's Formulas for Stress and Strain*, 7th ed. (McGraw-Hill, 2002) (S9, S10, S11).
- Physical constants and material properties (Stefan-Boltzmann constant, water expansion on freezing 9.17 %, surface tension of water 0.072 N/m, 6063-T5 minimum yield about 110 MPa, typical 145 MPa) are handbook values; the 6063 yield is to be replaced with the stockist's certificate.